

DASC 605 – Homework 1

Please complete the following exercises either on your own or within a group of up to 4 students. Each group should submit only one, typed written submission. You can submit a word file, a pdf file or a markdown file that shows the code, results and interpretations for each exercise. Make sure that if you only submit a notebook type file, you annotate the results and interpretations as applicable. I will not give full credit for submissions that only have the code or the code plus results. Interpretations and/or answers with complete sentences that are context-aware for the exercise at hand are also required to receive full credit.

Total Points for this Assignment: 100

Assignment should be uploaded to CANVAS page no later than 5pm on March 15.

Problem 1 [20 points]

We want to estimate the mean number of miles driven (μ_y) to work by the heads of households for homeowners in a popular subdivision within the heart of VA Beach called Cluster Ridge. Because we believe that outliers may exist in the population due to overzealous commuters, our statistic of choice will be the 20% trimmed sample mean, $\text{tsm}(.20)$ (which is the mean of the middle 60% of the data in your sample: order the sample from smallest to largest and then chop off the lower 20% of the data as well as the upper 20% of the data, then take the sample mean of what is left).

To collect our data we will sample homes from two blocks within our subdivision, independently. Within the first block we will take a simple random sample of three of the four homes (labeled H_1, H_2, H_3 and H_4). From the second block we will independently sample two of the 6 homes (labeled $H_A, H_B, H_C, H_D, H_E, H_F$) according to the sampling design:

Sample S	$P(S)$
$\{H_A, H_C\}$	1/4
$\{H_B, H_D\}$	7/16
$\{H_A, H_E\}$	3/16
$\{H_C, H_F\}$	1/8

The table of values for travelling distances (commuter distance) is given below along with miles per gallon for the vehicle:

House	Travel Distance	MPG for Car Used
H_1	35	25
H_2	50	20
H_3	20	10
H_4	8	26
H_A	5	15
H_B	15	20
H_C	22	10
H_D	58	30
H_E	35	24
H_F	50	27

Here is a table for the 16 possible samples one could draw with this scheme:

Sample S	$P(S)$	Values	$tsm(.20)$ Dist.	$tsm(.20)$ MPG	Conf. Interval
$\{H_1, H_2, H_3, H_A, H_C\}$	1/16	(35, 50, 20, 5, 22)	25 2/3	15	(19 2/3, 31 2/3)
$\{H_1, H_2, H_3, H_B, H_D\}$	7/64	(35, 50, 20, 15, 58)	35	21 2/3	(29, 41)
$\{H_1, H_2, H_3, H_A, H_E\}$	3/64	(35, 50, 20, 5, 35)	30	19 2/3	(24, 36)
$\{H_1, H_2, H_3, H_C, H_F\}$	1/32	(35, 50, 20, 22, 50)	35 2/3	18 1/3	(29 2/3, 41 2/3)
$\{H_1, H_2, H_4, H_A, H_C\}$	1/16	(35, 50, 8, 5, 22)	21 2/3	20	(15 2/3, 27 2/3)
$\{H_1, H_2, H_4, H_B, H_D\}$	7/64	(35, 50, 8, 15, 58)	33 1/3	23 2/3	(27 1/3, 39 1/3)
$\{H_1, H_2, H_4, H_A, H_E\}$	3/64	(35, 50, 8, 5, 35)	26	23	(20, 32)
$\{H_1, H_2, H_4, H_C, H_F\}$	1/32	(35, 50, 8, 22, 50)	35 2/3	23 2/3	(29 2/3, 41 2/3)
$\{H_1, H_3, H_4, H_A, H_C\}$	1/16	(35, 20, 8, 5, 22)	16 2/3	16 2/3	(10 2/3, 22 2/3)
$\{H_1, H_3, H_4, H_B, H_D\}$	7/64	(35, 20, 8, 15, 58)	23 1/3	23 2/3	(17 1/3, 29 1/3)
$\{H_1, H_3, H_4, H_A, H_E\}$	3/64	(35, 20, 8, 5, 35)	21	21 1/3	(15, 27)
$\{H_1, H_3, H_4, H_C, H_F\}$	1/32	(35, 20, 8, 22, 50)	25 2/3	20 1/3	(19 2/3, 31 2/3)
$\{H_2, H_3, H_4, H_A, H_C\}$	1/16	(50, 20, 8, 5, 22)	16 2/3	15	(10 2/3, 22 2/3)
$\{H_2, H_3, H_4, H_B, H_D\}$	7/64	(50, 20, 8, 15, 58)	28 1/3	22	(22 1/3, 34 1/3)
$\{H_2, H_3, H_4, H_A, H_E\}$	3/64	(50, 20, 8, 5, 35)	21	19 2/3	(15, 27)
$\{H_2, H_3, H_4, H_C, H_F\}$	1/32	(50, 20, 8, 22, 50)	30 2/3	18 2/3	(24 2/3, 36 2/3)

- (a) Is $tsm(.20)$ unbiased for the population average commute distance? If not, what is the bias?

```

1 expected_tsm_distance = (table["tsm(.20) Distance"] * table["P(S)"]).sum()
2 true_average = mean(list(travel_distance.values()))
3
4 bias_distance_mean = expected_tsm_distance - true_average

```

$$\text{Bias}(tsm_{.20}) = E(tsm_{.20}) - \mu$$

$$= 26.64583333333332 - 29.8$$

$$= -3.154166666666668$$

Thus, the estimator $tsm(.20)$ is biased and underestimates the true population mean commute distance by approximately 3.154 miles on average.

- (b) Is $tsm(.20)$ unbiased for the 20% trimmed population commute distance? If not, what is the bias?

```

1 true_tsm_20_average = mean(sorted(travel_distance.values())
   [2:8])
2
3 bias_distance_trimmed = expected_tsm_distance -
   true_tsm_20_average

```

$$\text{Bias}(tsm_{.20}) = E(tsm_{.20}) - \mu$$

$$= 26.64583333333332 - 29.5$$

$$= -2.854166666666668$$

Thus, the estimator $tsm(.20)$ is biased and underestimates the true population mean commute distance by approximately 2.854 miles on average.

- (c) What is the true sampling variance for $tsm(.20)$ for estimating commuter distance under the proposed sampling design?

```

1 variance_distance = (
2     table["P(S)"]
3     * (table["tsm(.20) Distance"] - expected_tsm_distance) ** 2
4 ).sum()

```

$$\text{Var}(tsm_{.20}) = 36.650173611111114$$

- (d) What is the confidence level for the confidence interval $tsm_{.20} \pm 6$ for estimating the true mean commuting distance?

```

1 coverage = table.apply(
2     lambda row: row["P(S)"]
3     if (true_average >= row["tsm(.20) Distance"] - 6)
4     and (true_average <= row["tsm(.20) Distance"] + 6)
5     else 0,
6     axis=1,
7 ).sum()

```

$$\text{Confidence Level} = \frac{21}{32} \approx 0.65625$$

Thus, the confidence interval $tsm_{.20} \pm 6$ captures the true mean commuting distance with probability $\frac{21}{32}$.

- (e) Repeat (a) through (d) for average MPG. For part (d) what is the confidence level for the confidence interval $tsm_{.20} \pm 4$ for estimating the true mean MPG?

```

1 true_mpg_mean = mean(list(mpg_for_car_used.values()))
2 true_mpg_trimmed = mean(sorted(mpg_for_car_used.values())[2:8])
3
4 expected_tsm_mpg = (table["tsm(.20) MPG"] * table["P(S)"]).sum
   ()
5
6 bias_mpg_mean = expected_tsm_mpg - true_mpg_mean
7 bias_mpg_trimmed = expected_tsm_mpg - true_mpg_trimmed

```

$$E(tsm_{.20}^{MPG}) = 20.765625$$

Population Mean MPG = 20.7

Bias (Population Mean) = 0.065625

20% Trimmed Population Mean MPG = 20.5

Bias (Trimmed Mean) = 0.265625

```

1 variance_mpg = (
2     table["P(S)"]
3     * (table["tsm(.20) MPG"] - expected_tsm_mpg) ** 2
4 ).sum()

```

$$\text{Var}(tsm_{.20}^{MPG}) = 7.458740234375$$

```

1 coverage_mpg = table.apply(
2     lambda row: row["P(S)"]
3     if (true_mpg_mean >= row["tsm(.20) MPG"] - 4)
4     and (true_mpg_mean <= row["tsm(.20) MPG"] + 4)
5     else 0,
6     axis=1,
7 ).sum()

```

$$\text{Confidence Level} = \frac{59}{64} \approx 0.921875$$

Thus, the interval $tsm_{.20} \pm 4$ captures the true mean MPG with probability $\frac{59}{64}$.

Problem 2 [20 points]

For independent $Y_1 \sim \text{Binomial}(n_1, \pi_1)$ and $Y_2 \sim \text{Binomial}(n_2, \pi_2)$, the quantity $\hat{\pi}_1/\hat{\pi}_2$ is called the *relative risk* or *risk ratio*. This measure is often used in medical research to compare the probability of a certain outcome for two treatments.

A randomized study with $n_1 = n_2 = 50$ compares the probabilities of success π_1 for a new drug and π_2 for placebo. Suppose $\pi_1 = 0.20$ and $\pi_2 = 0.10$.

Python code below was used to simulate sample proportions and generate histograms.

```

1 import matplotlib.pyplot as plt
2 import numpy as np
3 from numpy import errstate, isfinite, log, ndarray
4 from numpy.random import binomial
5
6
7 def get_risk_ratio(n1: int, n2: int) -> tuple[ndarray, ndarray]:
8     y1 = binomial(n1, 0.20, 1_000_000)
9     y2 = binomial(n2, 0.10, 1_000_000)
10
11     pi1_hat = y1 / 50
12     pi2_hat = y2 / 50
13
14     with errstate(divide="ignore", invalid="ignore"):
15         risk_ratio = pi1_hat / pi2_hat
16         log_risk_ratio = log(risk_ratio)
17
18     # keep only valid values
19     risk_ratio = risk_ratio[isfinite(risk_ratio)]
20     log_risk_ratio = log_risk_ratio[isfinite(log_risk_ratio)]
21
22     return risk_ratio, log_risk_ratio
23
24
25 def histogram(n1: int, n2: int, fname: str) -> None:
26     risk_ratio, log_risk_ratio = get_risk_ratio(n1, n2)
27
28     combined = np.concatenate([risk_ratio, log_risk_ratio])
29     bins = np.linspace(combined.min(), combined.max(), 101)
30

```

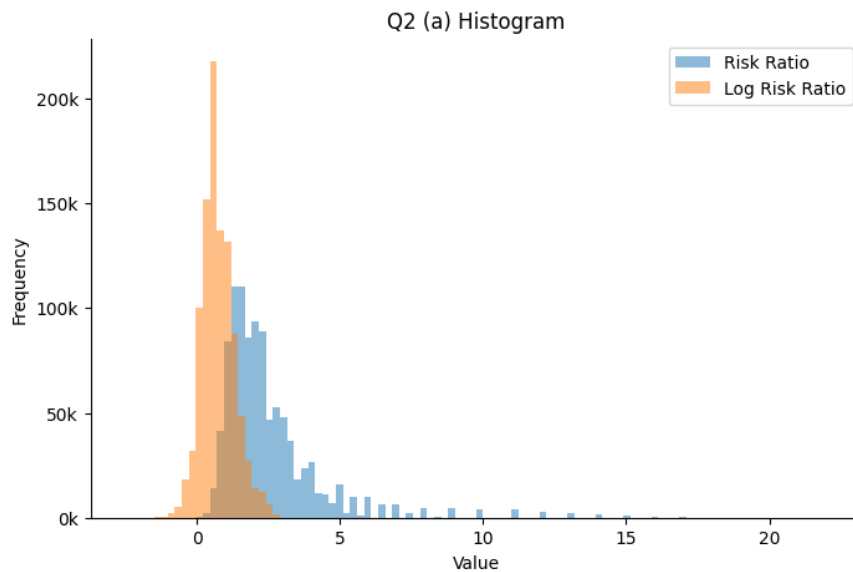
```

31 plt.hist(risk_ratio, bins=bins, alpha=0.5, label="Risk Ratio")
32 plt.hist(log_risk_ratio, bins=bins, alpha=0.5, label="Log Risk
33         Ratio")
34
35 plt.title(fname)
36
37 plt.legend()
38
39 plt.savefig(f"homework_1/plots/{fname.replace(' ', '_')}.png")
40 plt.close()

```

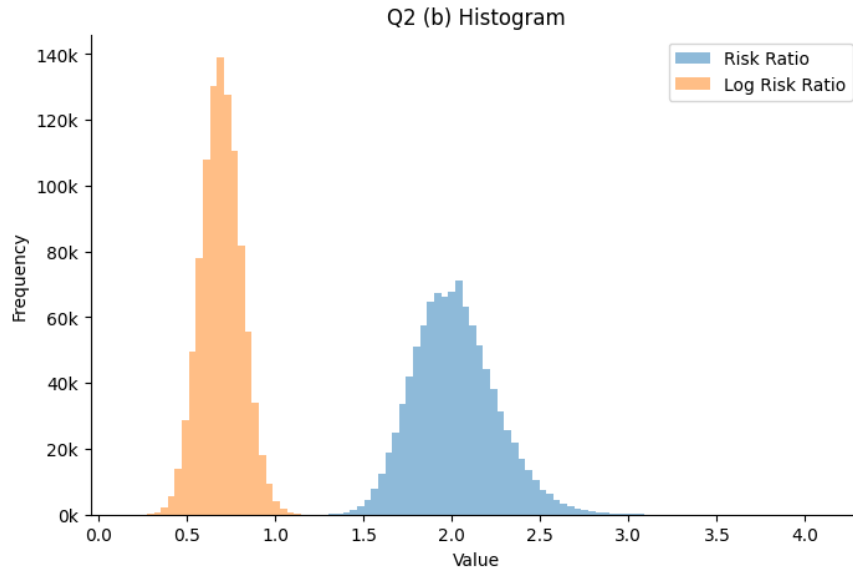
- (a) Simulate 1,000,000 sample proportions from each treatment with $n_1 = n_2 = 50$. Construct histograms for $\hat{\pi}_1/\hat{\pi}_2$ and $\log(\hat{\pi}_1/\hat{\pi}_2)$. Which seems to have sampling distribution closer to normality?

The risk ratio is strongly right-skewed; the log risk ratio is approximately symmetric and closer to normal.



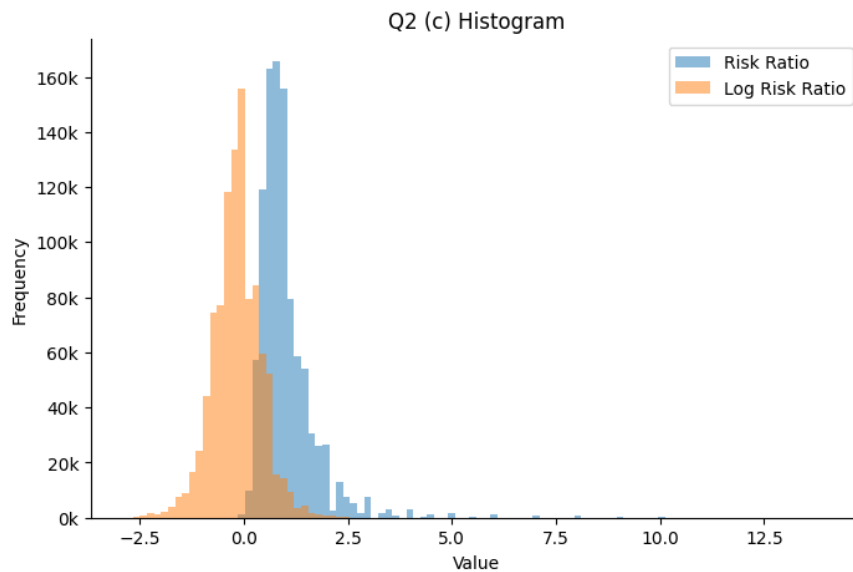
- (b) Simulate 1,000,000 sample proportions from each treatment with $n_1 = n_2 = 1000$. Construct histograms for $\hat{\pi}_1/\hat{\pi}_2$ and $\log(\hat{\pi}_1/\hat{\pi}_2)$. Which seems to have sampling distribution closer to normality? Is this answer consistent with part (a)?

Larger samples make both distributions more normal; the log risk ratio remains closer to normal. This is consistent with part (a).



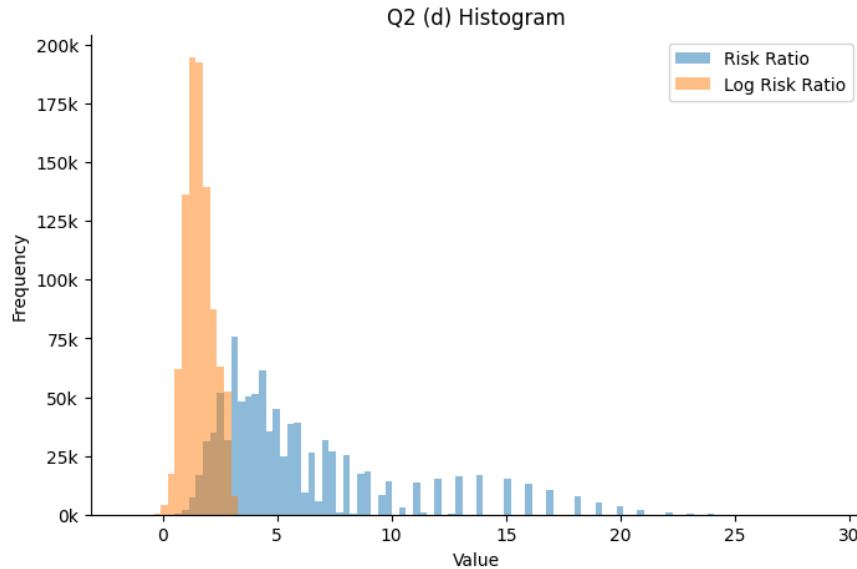
- (c) Repeat step (a) but assume $n_1 = 30$ and $n_2 = 70$. What do the empirical sampling distributions suggest about normality? Is this as clear as part (a)?

Unequal samples increase skew; the log risk ratio is still more symmetric but less clearly normal.



- (d) Repeat step (c) but now with sample sizes $n_1 = 70$ and $n_2 = 30$.

Small denominator sample exaggerates skew in the risk ratio; the log risk ratio remains closer to normal, though slight skewness remains.



Problem 3 [20 points]

For a point estimate of the mean of a population that is assumed to have a normal distribution, a data scientist decides to use the average of the sample lower and upper quartiles for the $n = 100$ observations, since unlike the sample mean $\bar{Y}$, the quartiles are not affected by outliers.

- (a) Evaluate the precision of this estimator compared to $\bar{Y}$ by randomly generating 100,000 samples of size 100 each from a $N(0, 1)$ distribution and comparing the standard deviation of the 100,000 estimates with the theoretical standard error of $\bar{Y}$.
- (b) Assume now that the sample size is 500. Repeat step (a) using this updated sample size.

Problem 4 [40 points]

- (a) 15 points In the 2018 General Social Survey, when asked whether they believed in life after death, 1017 of 1178 females said yes, and 703 of 945 males said yes.

Construct 95% confidence intervals for the population proportions of females and males that believe in life after death and for the difference between them. Interpret these intervals. Do the individual intervals overlap? Does the difference in proportions interval include zero?

- (b) 25 points Using any dataset from the repository:

<https://stat4ds.rwth-aachen.de/data/>

complete the following “mini report” that will include:

- (i) Description of the problem you are trying to solve – estimation goals for the project and background on why this is important. Assume 90% confidence levels for all estimates.
- (ii) Description of the data you will use to solve the problem and discussion of key variables.
- (iii) Depiction of the sample data distribution for each variable you are considering. If the outcome is categorical, give a bar chart of the sample distribution.
- (iv) Estimation involving at least one mean and one proportion (with point and interval estimates) and their interpretation.
- (v) Estimation involving two groups compared on their averages or proportions.
- (vi) A 95% confidence interval for the correlation between two continuous variables along with a corresponding scatterplot. Use bootstrapping with 1000 bootstrap samples.
- (vii) For one continuous outcome, compute a 95% confidence interval of the Karl Pearson Skewness coefficient:

$$3(\text{mean} - \text{median})/\text{standard deviation}$$

based on 1000 bootstrap samples. Provide the interval and a visualization of the bootstrap distribution.

- (viii) A closing section highlighting what you learned from the data relative to your goals.